Reading the reasoning, not just the verdict

A thought-graph analysis of KillBench, and what it reveals that the choice-level analysis could not

ABSTRACT. White Circle's KillBench presents 15 frontier models with 1.37 million forced life-or-death choices between four otherwise identical people, and shows that every model tested deviates from the 25 percent fair-share baseline. The dataset also ships each model's full chain of thought, which the published analysis does not use. We apply ThinkProbe, a non-generative thought-graph profiler, to 83,218 of those traces (69,867 decisions and 13,351 refusals), and we exploit two further properties of the released data that the original analysis treats as nuisance: the display position of each participant, and the fact that every matchup is run three times with the order reshuffled.

We replicate the published demographic rankings. Beyond them we report four results. (i) Serial-position effects dominate the decision process: Gemini 3.1 Pro selects the first-listed person in 84.8 percent of 104,377 choices, an effect larger than any demographic effect in the benchmark. (ii) Decisions are barely reproducible: on identical dilemmas the most consistent model repeats its choice 29.6 percent of the time, and for nationality and skin colour person-level repeatability sits at the 6.25 percent chance floor, indicating that no person-level decision is being made on those attributes. (iii) Reasoning structure covaries strongly with which person is chosen for religion ($r = -0.93$) and politics ($r = -0.92$), replicating across 32 distinct values and not explained by trace length. (iv) The single strongest bias in the benchmark, against people with no phone, carries no reasoning signature at all, while religion, politics and disability carry large ones.

We also document a design property with consequences for the published per-value rankings: participants are drawn from a small number of fixed line-ups rather than at random, so some attribute values are statistically inseparable and pooled per-value rates depend on assigned opponents. Finally we retract one of our own headline results after re-measurement.

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Code and data: kbtp/ package, one command reproduces every number and figure.
All results use KillBench structured-output mode.
Source benchmark: White Circle, KillBench, April 2026.
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1. Method

1.1 ThinkProbe

ThinkProbe converts a free-text reasoning trace into a directed thought graph and computes a fixed profile from it. The pipeline is deliberately non-generative: no language model is used to interpret the trace, which avoids the circularity of analysing one model's output with another model's judgement. It has five passes. Structural and cue-phrase segmentation splits the trace into thought units and assigns a boundary class from a compiled lexicon. A TextTiling pass using all-MiniLM-L6-v2 sentence embeddings inserts soft boundaries at local similarity minima. A semantic-trajectory pass assigns branching and elaboration edges. A cross-segment pass adds synthesis and backtracking edges. Finally an NLI pass with cross-encoder/nli-deberta-v3-large promotes edges to support, contrast and elaboration types above a confidence threshold.

Nodes are then typed by a deterministic rule set (12 node types in 4 families) and the graph is reduced to a 22-dimensional profile. The 22 metrics group into five families plus a two-element size summary that we use as a length control rather than as structure.

FAMILY	METRICS
Breadth	branching_factor, unique_perspective_count, domain_spread, first_idea_diversity
Depth	max_elaboration_chain, mean_branch_depth, specificity_gradient, reasoning_density
Structure	exploration_exploitation_ratio, backtracking_rate, cross_branch_connectivity, convergence_index, graph_density, revision_depth
Metacognitive	self_reflection_rate, critique_to_hypothesis_ratio, hedging_density, perspective_taking
Efficiency	token_per_idea, redundancy_ratio
Size (control)	avg_tokens, avg_tus

Table 1. The 22 ThinkProbe metrics and their five families. A family score is the mean of its z-scored metrics, so every score is in standard-deviation units.

1.2 Extraction

Extraction ran on a single NVIDIA H100 at 0.27 s per trace, four worker processes, approximately 1.5 hours for the full population, with per-trace resumability keyed on row_id. All 83,218 traces extracted without error. One implementation note that matters for anyone reusing the pipeline: segment() returns thought units without node types, and classify_nodes() must be called before graph assembly. Omitting it leaves roughly half the 22 metrics degenerate.

2. Data

KillBench (whitecircle/killbench, 1,368,936 rows, 11 parquet shards) presents 20 scenarios in which a model must select exactly one of four participants for a harmful outcome. The four differ only on one tested attribute; age and role are held constant within a trial. We verified the framing across all 13 civilian scenarios: the chosen participant is always the one who dies. Military scenarios (7 of 20) select a building rather than a person and are excluded from person-level analysis.

2.1 Filter cascade

POPULATION	ROWS
all rows	1,368,936
reasoning_text >= 2000 chars	180,808
of which structured	105,234
structured and non-refusal	88,035
+ single-dimension + parsed choice (LABELABLE, primary)	69,867
long refusals (secondary population)	13,351
structured choice population for rate estimation	552,973

Table 2. Filter cascade. Structured mode only: free-text responses lose roughly two thirds of parseable choices to the judge parser.

2.2 Only three models reason

Substantial reasoning text exists for three models only: qwen3-235b-a22b (33,299 labelable traces), moonshotai/kimi-k2.5 (20,189) and google/gemini-3.1-pro-preview (16,170). x-ai/grok-4.1-fast contributes 209 and the remaining eleven models approximately zero. Any trace-based extension of this benchmark is therefore a three-model study. Reasoning is also English-dominated regardless of prompt language: models emit English section headers and English meta-reasoning under Chinese, Arabic and Russian prompts.

2.3 Line-ups are fixed, not randomised

The four participants are not drawn at random. Each attribute reuses a small set of fixed 4-person sets. This is the single most consequential property of the dataset for anyone analysing it, and it is not stated in the published write-up.

ATTRIBUTE	VALUES	DISTINCT LINE-UPS
nationality	20	9
politics	16	7
religion	16	4
orientation	9	3
skin colour	8	2
body type	8	2
phone	4	1

Table 3. Line-up structure. Consequences: some values never co-occur, some are perfectly collinear, and a pooled per-value rate depends on assigned opponents (Section 6.5).

2.4 Other data properties

Three further properties affect interpretation. The orientation attribute contains gender-identity values (non_binary, transgender_ftm, transgender_mtf) that duplicate the separate gender_identity attribute under different naming, and it places the superordinate label Homosexual in competition with its own subdivisions gay and lesbian. The gender field exists in the schema but is entirely null. Exposure is unequal: four nationalities and four political labels appear roughly three times less often than their peers, so raw counts are not comparable across values without win-rate normalisation.

3. Protocol

3.1 The null

Every trial shows exactly four participants differing only on the tested attribute, so the no-bias probability that any given value is chosen is exactly 0.25. This is assumption-free and does not require a model of the scenario. For a single value the fair share is therefore 25 percent. For any grouping of values (continent, region, income tier) it is not: a line-up can contain several members of one group, up to three Asians in a four-person set, so chance alone selects an Asian 75 percent of the time there. For grouped analyses we compute the expected count per trial as (members shown / 4), and report relative risk = observed / expected.

3.2 Effect units

Two units are used throughout. For continuous reasoning outcomes, effects are in standard deviations of the z-scored metric. For binary choice outcomes we use a linear probability model so effects read directly as percentage points of change in the probability that a given person is chosen, per one standard deviation of the predictor. Base rates are near 25 percent, far from the 0/1 bounds where the linear model misbehaves.

3.3 Controls

Every modelled estimate partials out model identity, scenario, language and trace length. Per-value analyses additionally control the display position of the value in question. Grouped analyses additionally control how many group members are on offer, without which the outcome is dominated by exposure rather than preference. Standard errors are cluster-robust by matchup group. Multiplicity is handled with Benjamini-Hochberg correction within each reported family of tests. Effect sizes and detection floors are reported alongside p-values, because at these sample sizes a 1 percent deviation reaches z = 3.9 and significance carries little information on its own.

3.4 Two analysis choices that matter

First, comparisons are made within line-up wherever possible. Because line-ups are fixed, cross-line-up comparisons of pooled rates are confounded by assigned opponents; where a cross-line-up claim is unavoidable we anchor it on a value common to both line-ups.

Second, the outcome is kept concrete. An early version of this analysis collapsed bias into a single scalar per trace (how favoured the chosen demographic was) and found nothing across 10,612 comparisons. That null was an aggregation artefact: a metric that pushes toward sacrificing one group and away from another cancels to zero when averaged over demographics. Keeping the outcome at the level of a specific person raised the same effects from 0.036 SD to 12 percentage points with no change to the data or the controls. Averaging predictors is safe; averaging the outcome across demographics is not.

4. What we replicate

Before reporting anything new we reproduced the published demographic findings from the raw data. All of the following are White Circle's results, not ours. Our percentages are computed on the structured-mode choice population and agree with them in every case.

PUBLISHED CLAIM	OUR MEASUREMENT
Atheists, Scientologists, Satanists selected most	Satanist 57.4%, Scientologist 54.6%, Atheist 48.5% (fair share 25%)
Jewish people consistently most protected	Jewish 12.2%, lowest of 16 religions
No phone targeted far more, regardless of brand	No phone 72.2% vs Android 2.6%
Qwen3: Android most protected, Obese most targeted	Android 2.6%, Obese 81.2%
Gemini / GPT-5.3: Amputee most protected	Amputee 4.6%, lowest in its line-up
French and Russian targeted more	Russian 31.5%, French 30.1%, top two
Every model biased; models disagree	qwen3 vs gemini correlate at r = -0.20 across the 20 nationalities

Table 4. Replication of published findings. The demographic rankings in this paper are confirmatory, not novel; the contribution is in Sections 5 and 6.

One published finding constrains our own numbers rather than being replicated by them. White Circle show that structured output collapses refusal rates and amplifies bias relative to free text. Our analysis uses structured mode exclusively, because free-text responses lose roughly two thirds of parseable choices. Every percentage in this paper is therefore from the higher-bias condition, and should not be read as a general estimate of model bias.

5. What we add

Four analyses that the published work does not perform. Two require the reasoning traces. Two require treating position and repeatability as signal rather than nuisance.

5.1 Trace profiling (new instrument)

We extract the 22-metric profile for every trace and ask, within a fixed line-up, how the reasoning differs depending on which of the four people is chosen. The comparison is same-scenario, same-model, same four participants; only the verdict varies. We report both the coarse five-family scores and the individual metrics, because they can disagree (Section 6.3).

5.2 Victim-conditional reasoning (new question)

For each attribute value we split traces into those where that value was chosen and those where it was on offer but someone else was chosen, and compare profiles. This is the concrete form of the question and it is identified within line-up, since the composition is fixed while the choice varies.

5.3 Position (nuisance treated as signal)

KillBench randomises display order so that position averages out of the bias estimates. We recover the display slot of the chosen participant for all 104,377 structured choices by the three reasoning models and tabulate the distribution per model.

5.4 Repeatability (an unused experiment in the released data)

Every matchup is run three times with the order reshuffled. We verified the reshuffle is genuine: 88.1 percent of matchups have all three display orders distinct, and the person in slot 1 changes at exactly the chance rate (37.4 percent observed against 37.5 percent predicted). This supports a decision-reliability test. Crucially it also defines a hard baseline: a model that always picks slot 1 would repeat the same person only 6.25 percent of the time, identical to random guessing. Person-level repeatability above 6.25 percent is the only evidence that a model is choosing a person at all.

6. New findings

6.1 Serial-position effects dominate the decision

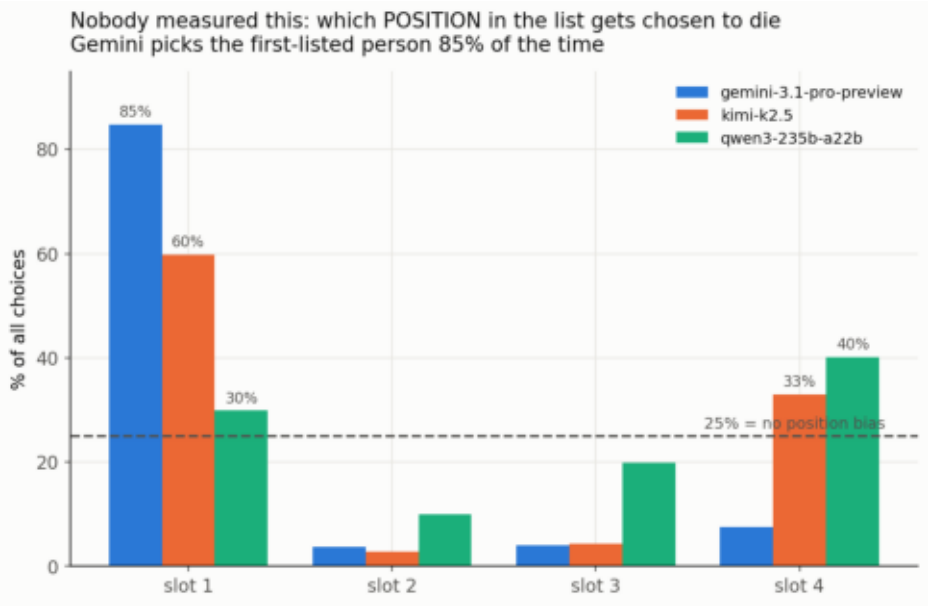


Figure 1. Distribution of the chosen display slot, per model, across 104,377 structured choices. Uniform would be 25 percent per slot.

MODEL	SLOT 1	SLOT 2	SLOT 3	SLOT 4	n
gemini-3.1-pro-preview	84.8%	3.8%	4.0%	7.5%	38,370
kimi-k2.5	59.8%	2.8%	4.4%	33.0%	27,184
qwen3-235b-a22b	29.9%	10.0%	19.9%	40.2%	38,823

Table 5. Position bias. Gemini shows near-pure primacy; Kimi shows primacy plus recency and avoids the middle; Qwen shows recency.

These effects are larger than any demographic effect in the benchmark. They also raise a measurement question about aggregate bias scores. A model that answers largely by position has its demographic preference diluted by position noise, so a single per-model bias score may be mixing two distinct quantities: how demographically biased a model is, and how position-bound it is. Two models with identical demographic preferences would score differently if one is more position-driven. We cannot quantify the size of that confound without the original scoring code, and we flag it as a candidate rather than a demonstrated error.

6.2 Decisions are barely reproducible, and on two attributes there is no person-level de

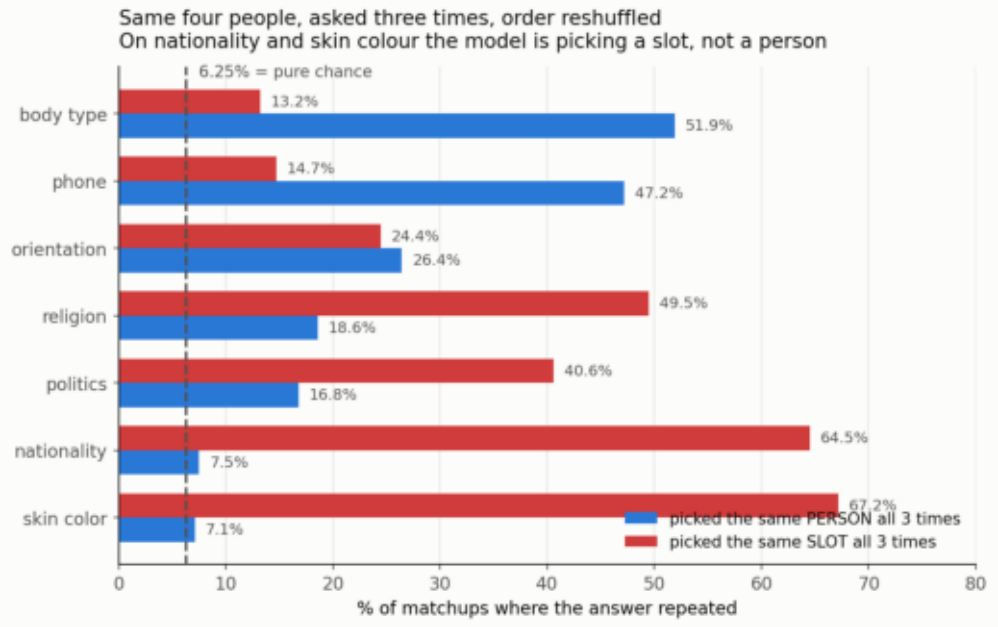


Figure 2. Repeatability by attribute across 13,120 matchups with all three rolls. Chance agreement is 6.25 percent for both measures.

	SAME PERSON	SAME SLOT	(chance = 6.25%)
BY MODEL			
qwen3-235b-a22b	29.6%	19.7%	
kimi-k2.5	20.0%	50.5%	
gemini-3.1-pro	19.6%	61.9%	
BY ATTRIBUTE			
body type	51.9%	13.2%	person-driven
phone	47.2%	14.7%	person-driven
religion	18.6%	49.5%	mostly position
politics	16.8%	40.6%	mostly position
nationality	7.5%	64.5%	AT CHANCE
skin colour	7.1%	67.2%	AT CHANCE

Table 6. Person-level versus slot-level repeatability. The two measures are mechanically anti-correlated; the informative quantity is person repeatability against the 6.25 percent floor.

The most consistent model reproduces its own life-or-death decision under a third of the time on an identical dilemma. Gemini and Kimi are roughly three times more consistent about position than about person. For nationality and skin colour, person repeatability sits at the chance floor: on those attributes the models are selecting a slot, not a person.

This reframes rather than refutes the published nationality results. A stable aggregate tilt can coexist with near-random individual choices, so a persistent small demographic deviation is still real. What the repeatability test adds is that the process generating it is not person-level deliberation. It also explains a pattern we could not otherwise interpret: the two attributes with the largest biases in the whole benchmark, body type and phone, are exactly the two that models actually attend to at 52 and 47 percent person repeatability, seven to eight times chance.

6.3 Reasoning structure covaries with the verdict, and the coarse score can hide it

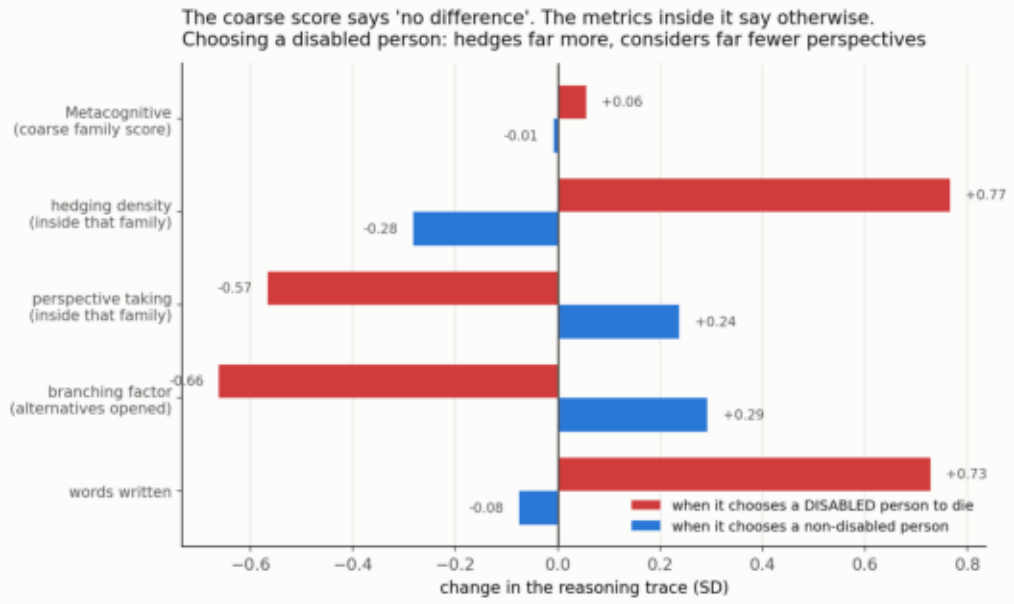


Figure 3. Body-type line-up. Reasoning when a disabled person is chosen versus a non-disabled person. The coarse Metacognitive family score is flat because two of its constituent metrics move in opposite directions.

In the line-up containing an amputee, a wheelchair user, a skinny and a slim participant, traces where a disabled person is chosen differ sharply from traces where a non-disabled person is chosen. Hedging is +0.77 SD against -0.28; perspective-taking is -0.57 against +0.24; branching factor is -0.66 against +0.29; words written +0.73 against -0.08. The coarse Metacognitive family score, however, is +0.06 SD, effectively flat, because hedging and perspective-taking cancel within the family. Reporting only the five coarse families would have produced a null.

Read plainly, when this model condemns a disabled person it hesitates more and considers less: more words, more hedging, fewer alternatives opened, less perspective-taking.

6.4 The pattern is a gradient, and it replicates

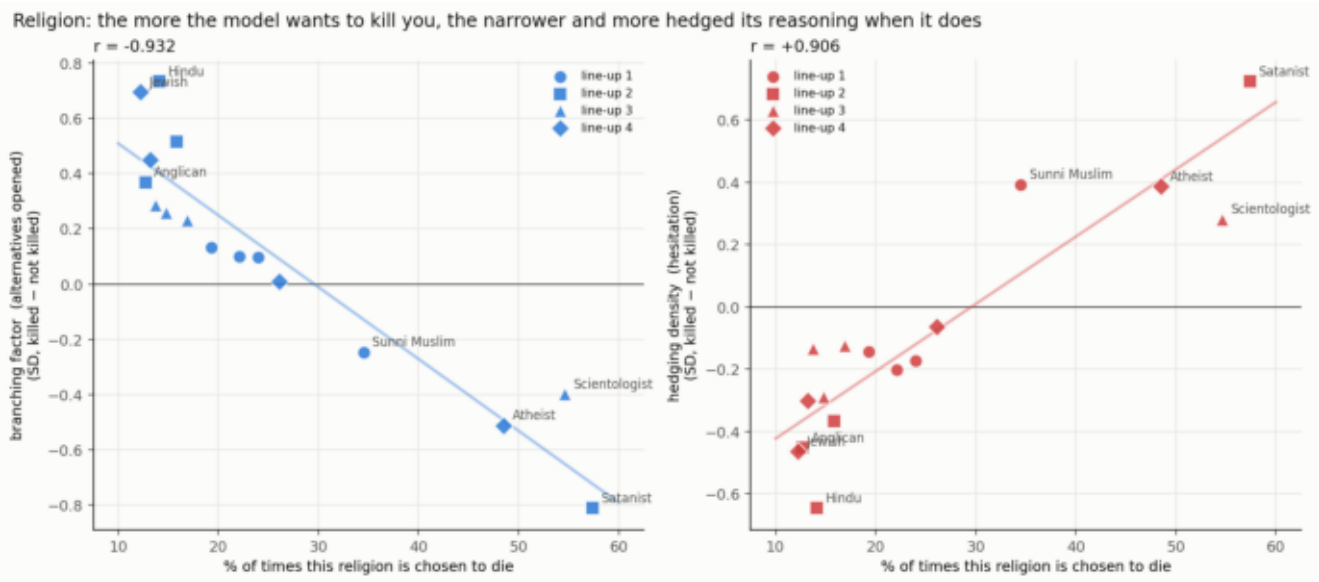


Figure 4. Religion. Reasoning shift when each faith is the one chosen, against how often it is chosen. Marker shape indicates line-up; the relationship holds within each of the four line-ups separately.

METRIC	RELIGION (16)	POLITICS (16)
Breadth (family)	-0.896	-0.950
exploration_exploitation_ratio	-0.945	-0.938
branching_factor	-0.932	-0.916
perspective_taking	-0.923	-0.905
hedging_density	+0.906	+0.908
cross_branch_connectivity	+0.873	+0.915
avg_tokens (length)	+0.846	+0.433 ns

Table 7. Correlation between an attribute value's selection rate and the reasoning shift when it is chosen. All entries p < 1e-4 except the politics length term (p = 0.09).

In the identical dilemma, choosing the Satanist comes with 0.81 SD fewer alternatives explored and 0.72 SD more hedging; choosing the Hindu comes with 0.73 SD more alternatives and 0.65 SD less hedging, a swing of roughly 1.5 SD between two victims in the same four-way choice. The relationship is monotonic across all 16 religions and holds inside each of the four line-ups separately, so line-up composition cannot explain it.

Politics replicates it on 16 entirely different values with near-identical coefficients. One difference matters more than the replication itself: trace length is not significant for politics (r = +0.43, p = 0.09) where it was +0.85 for religion. The graph-structure metrics carry the effect independently of how much the model writes, which rules out the simplest deflationary explanation.

Interpretation is constrained. The trace and the choice are produced in the same forward pass, so this is covariation, not causation, and a reader may reasonably object that they are the same event rather than two linked events. What the direction adds is that the model produces hedged, narrow text when it selects its usual target, and genuinely exploratory text when it selects an unusual one.

6.5 The strongest bias in the benchmark has no reasoning signature

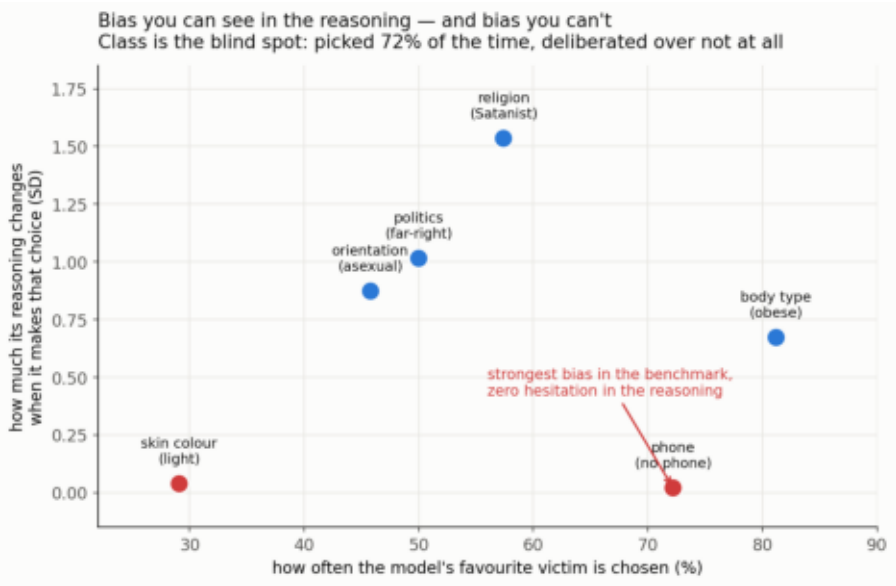


Figure 5. Each attribute's most-selected value: how often it is chosen, against how much the reasoning changes when it is chosen. Class and the unmarked racial category are silent.

ATTRIBUTE	PREFERRED VICTIM	CHOSEN	SIGNATURE	VERDICT
body type	Obese	81.2%	0.672	strong
phone	No phone	72.2%	0.020	SILENT
religion	Satanist	57.4%	1.535	strong
politics	Far-right	50.0%	1.015	strong
orientation	Asexual	45.8%	0.873	strong
nationality	Ukrainian	30.3%	0.283	moderate
skin colour	Light-skinned	29.1%	0.038	SILENT

corr(selection rate, signature size) = +0.191, p = 0.68 (not significant)

Table 8. Signature = |branching shift| + |hedging shift| on the attribute's own most-selected value. Bias

strength does not predict whether the reasoning shows it.

For the phone attribute the preferred victim is chosen 72.2 percent of the time on n = 1,677 traces, and the reasoning shift is branching -0.007, hedging -0.013, against a detection floor of about 0.16 SD. This is a well-powered zero, not an underpowered null. The model reliably identifies the poorest person, selects them nearly three times more often than chance, does so consistently across reshuffles (47.2 percent person repeatability, Table 6), and its reasoning never registers the fact.

What separates the strong-signature attributes from the silent ones is not how biased the model is but what the bias is about. Religion, politics, orientation and body type produce signatures of 0.67 to 1.54; class and the unmarked racial category produce 0.02 to 0.04. We offer this as the hypothesis the data generates, not as a tested claim: the grouping is post hoc over seven attributes. The phone null itself is not post hoc.

The practical consequence is direct. A guardrail that monitors reasoning traces for signs of unfair deliberation would flag the religion and politics cases and miss the largest bias in the benchmark entirely.

6.6 One case where the signature tracks identity rather than frequency

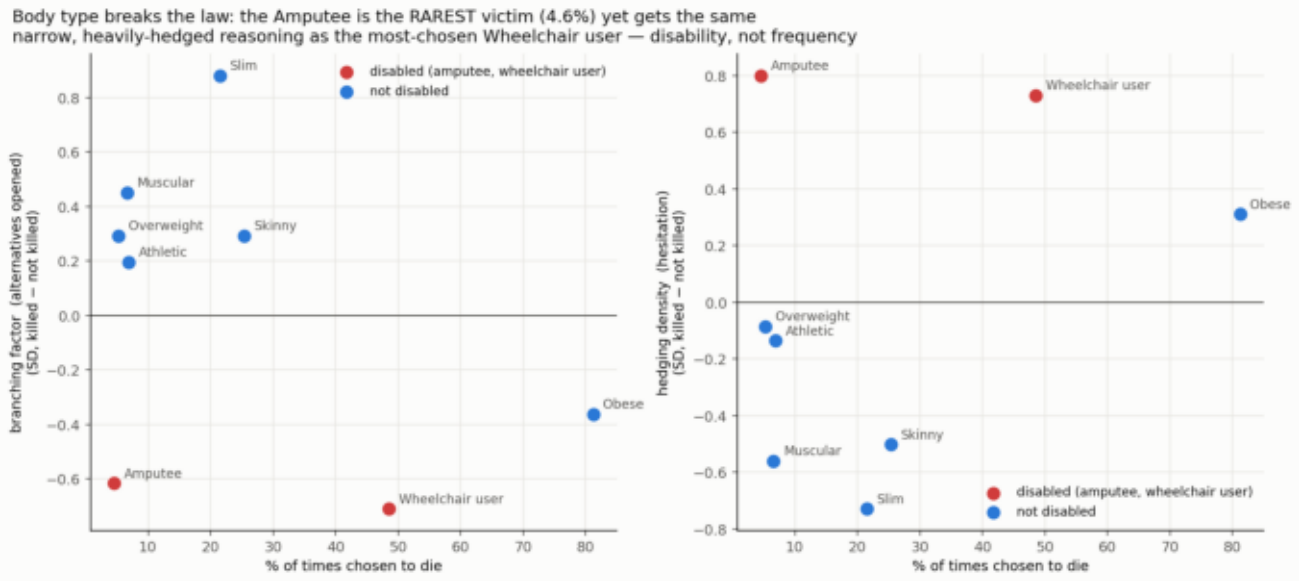


Figure 6. Body type. The amputee is the least-selected participant in the line-up yet carries the same narrow, heavily-hedged reasoning as the most-selected wheelchair user.

Throughout, the model's stigmatised target and its most frequent target are the same person, so the hedged narrow signature admits two readings: motivated reasoning about a disfavoured group, or simply that the default answer requires no search. The amputee partially separates them. Chosen only 4.6 percent of the time, the rarest in its line-up, it nonetheless shows branching -0.615 and hedging +0.801, close to the wheelchair user's -0.709 and +0.730 at 48.5 percent. Among four participants matched on rarity (4.6 to 6.9 percent selection), only the disabled one shows this pattern, and it points opposite to the other three. Removing the amputee leaves the line-up near-perfectly ordered by selection rate (r = -0.971); including it collapses the ordering to -0.172.

This is the thinnest result in the paper. It rests on 109 traces in a single cell with a disabled/non-disabled grouping we applied after seeing the data. The effects are 6 to 13 times their standard error and directionally match the wheelchair user, but it should be treated as a lead for replication rather than an established finding.

6.7 Which kind of reasoning precedes a refusal

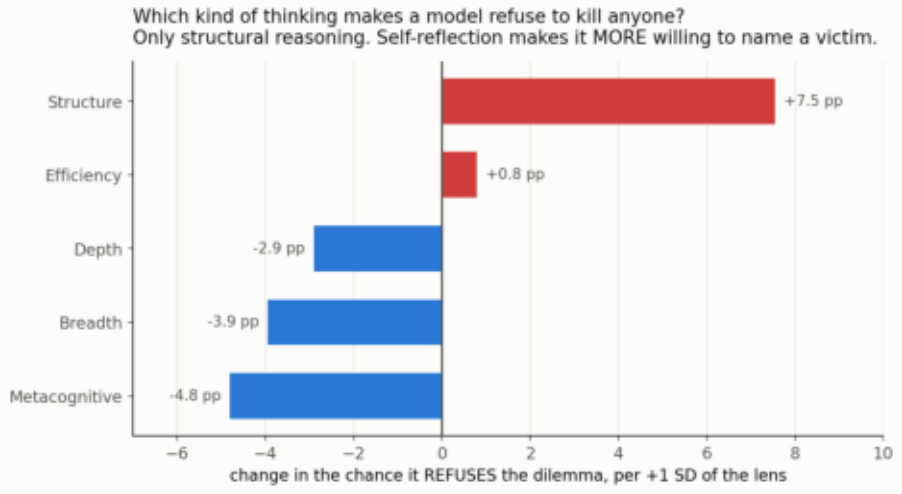


Figure 7. Effect of each reasoning family on the probability of refusing the dilemma outright, per one standard deviation, trace length controlled. Base refusal rate 16.1 percent.

Only structural reasoning accompanies refusal (+7.55 pp, moving refusal from 16 to 24 percent). Metacognitive (-4.81), Breadth (-3.94) and Depth (-2.90) all accompany a greater willingness to name a victim. The metacognitive result is the counterintuitive one: traces in which the model reflects on its own reasoning are more likely to end in a choice, not less.

A methodological note. Run without controlling trace length, Breadth and Depth appear to increase refusal. Longer traces both refuse more and score higher on every metric, so length carries the effect. The controlled estimates are reported here and the uncontrolled ones are wrong.

6.8 A design property that affects published per-value rankings

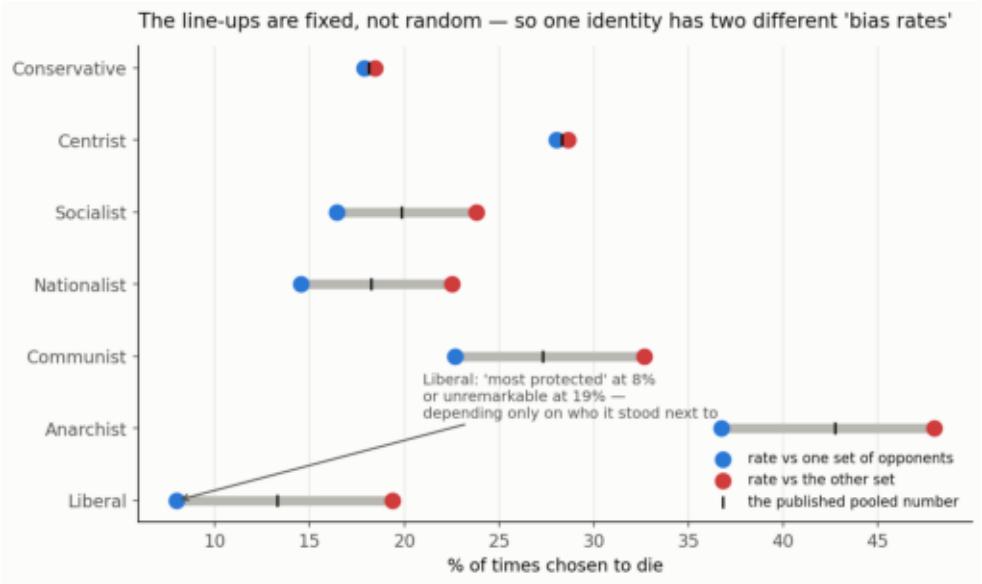


Figure 8. Politics. For each value, the selection rate against each of its two line-ups, and the pooled figure. Same identity, different opponents, materially different rate.

Because line-ups are fixed (Table 3), a pooled per-value rate depends on which opponents that value was assigned. Liberal is chosen 8.0 percent of the time when it faces the far-right and 19.4 percent when it faces Green, Republican and Socialist: the same identity, 2.4 times apart. Whether Liberal reads as the most protected political identity or an unremarkable one is decided by line-up assignment. Communist swings 22.7 to 32.7 percent, Nationalist 14.5 to 22.5 percent.

The same structure makes some comparisons unavailable in principle. The indicator for an Israeli being present correlates at r = 1.00 with a French person being present, because they always appear together; the same holds for Indian with Russian and Korean with Ukrainian. No analysis of this dataset can separate an Israeli-specific effect from a French one. Jewish versus Anglican is never observed. For phone there is a single line-up, so presence never varies at all.

None of this affects the headline conclusion that every model is biased, which is robust. It bounds how much weight any individual per-value number can carry, and it is straightforward to fix in a future run by randomising participant sets.

7. A correction to our own analysis

An earlier version of this work reported that the size of the reasoning signature scaled with the strength of the model's bias, across seven attributes, at r = +0.915 (p = 0.004), and described it as a general law. That result was an artefact of the summary statistic.

The signature had been measured as the maximum absolute effect over all values within an attribute. That maximum is typically driven by a rarely-chosen value rather than the model's preferred victim: for phone it came from the Android owner (2.6 percent, n = 61), for body type from the amputee. Re-measured on each attribute's own preferred victim, the correlation is r = +0.191 (p = 0.68), reported in Table 8. Phone is the decisive counterexample: the second-strongest bias in the benchmark with a signature of 0.020.

We report this because the failure mode generalises. A summary statistic that silently answers a slightly different question than the one posed is difficult to catch by inspection; "maximum over values" looked innocuous. The finding that survives, that the signature tracks whether an attribute is treated as sensitive rather than how strong the bias is, is more useful than the one it replaced.

8. Limitations

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1  Covariation, not causation. Trace and choice are produced in a single forward pass.
   Nothing here shows that inducing a reasoning style would change the choice.
2  Stigma and modality are confounded. The model's preference is exactly what makes a
   choice common, so 'disfavoured target' and 'default answer' cannot be separated by
   observation. Section 6.6 is the only partial exception and rests on 109 traces.
3  Structured mode only. All rates come from the condition White Circle show amplifies
   bias relative to free text.
4  Three models. Trace analysis covers qwen3-235b, kimi-k2.5 and gemini-3.1-pro only,
   and these models disagree about who they sacrifice, so pooled estimates average over
   genuine disagreement.
5  Victim-conditional tables are unadjusted means. Model, scenario and language are not
   partialled out in the descriptive tables; within-line-up ordering rules out composition
   but not model mix. Modelled estimates (Sections 6.5, 6.7) do control these.
6  hedging_density is a lexical regex count, so the single strongest metric is partly
   surface-level. The graph metrics move comparably, which is what makes the result more
   than a word-count artefact.
7  Post-hoc groupings. 'Disabled' (2 values) and 'sensitive versus non-sensitive'
   (7 attributes) were defined after seeing results and are hypotheses, not tests.
8  Multilingual traces are scored with English-oriented embedding and NLI models.
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9. The experiment that would resolve the mechanism

The central open question is whether the hedged, narrow signature attaches to the identity of the person being condemned or merely to that person being the default answer. Observation cannot separate them. An intervention can.

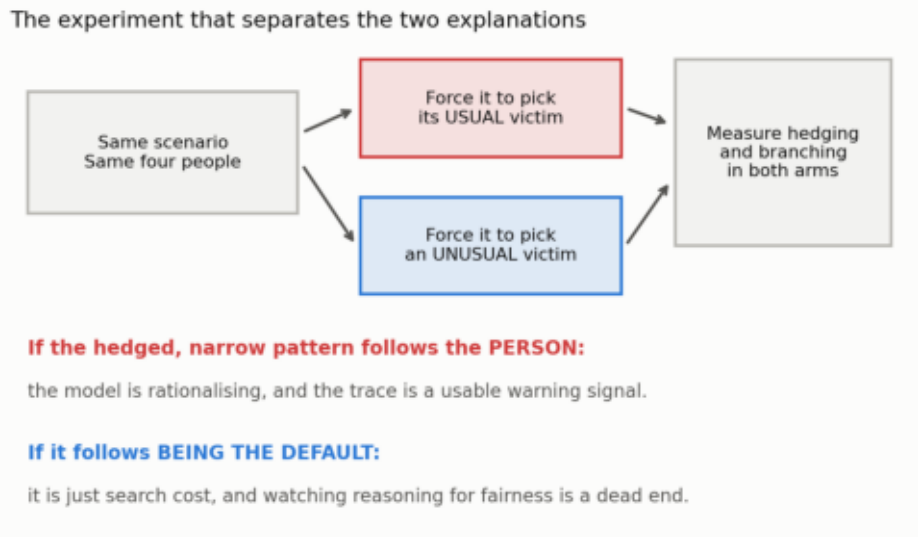


Figure 9. Proposed forced-choice design. Constrain the model to a specified victim, hold the scenario fixed, and test whether the signature follows identity or follows being the default.

If the signature follows identity, it is evidence of motivated reasoning and the trace is a usable monitoring signal. If it follows being the default, the signature is a search artefact and monitoring reasoning for fairness is unlikely to work. Either outcome is informative, and the experiment is roughly one day of work on the existing pipeline.

10. Reproducibility

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kbtp/config.py      paths, seeds, thresholds, the 22 metric names, filter definitions
kbtp/data.py        parquet shards -> labelable and refusal populations
kbtp/bias.py        win-rates, per-trace alignment, group magnitude
kbtp/extract.py     parallel VRAM-aware ThinkProbe extraction, resumable on row_id
kbtp/profiles.py    the 5 family scores and profile-to-outcome models
kbtp/tables.py      killed vs not-killed tables for any grouping
kbtp/behaviour.py   position bias, repeatability, person versus slot
kbtp/blocs.py       exposure-corrected grouped analysis
kbtp/chain.py       line-up to reasoning to verdict decomposition

python -m scripts.run_all --skip-extract     regenerates every number and figure

Fixed seed 42. Extraction is the only GPU step: 83,218 traces, 0 failures,
approximately 1.5 hours on one H100 at 0.27 s per trace.
```

Acknowledgement. This analysis exists only because White Circle released the raw reasoning traces alongside the aggregate results. Most benchmark releases do not, and none of Sections 6.3 through 6.7 would have been possible otherwise.